

Congruent Dissections of the Regular Tetrahedron: A Computational Atlas and Algebraic Synthesis

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Abstract

We give an exact computational atlas for the eight known congruent dissections of the regular tetrahedron T into $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ pieces. Each representative is reconstructed from canonical coordinates, its Dehn invariant is verified symbolically, and its refinement relations are recorded.

Every atlas dissection is G_{diss} -transitive, so Lagrange forces $n \mid 24$ for the known transitive family; full T_d -invariance occurs only for $n \in \{1, 12, 24\}$. Burnside-ring, 24-cell, and IFS-grammar models realise every divisor and provide a tested screening stack for related T_d -invariant orbit decompositions.

Beyond the atlas, we record two further scoped results. For $n = 16$ an all-one-face case is excluded under two finite MILP hypotheses, automatic in the Coxeter-pure instance. For $n = 5$, we prove a conditional obstruction under an explicit boundary-role homogeneity hypothesis (BRH): five congruent convex pieces cannot partition the regular tetrahedron when the same facets of the reference piece are exposed in every copy. The unrestricted case with different boundary roles remains open. We also record exact real-area bookkeeping and a chart-level algebraic check; these are not promoted to a general impossibility theorem. The new $n = 5$ computations use exact rational or symbolic arithmetic and are reproducible from the repository scripts.

Keywords: Regular tetrahedron, congruent dissection, Dehn invariant, tetrahedral symmetry group, Burnside ring, 24-cell, A_3 Coxeter complex, Divisor Completeness conjecture, Transitivity conjecture, face-to-face tiling.

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1 Introduction and summary of results

The dissection of the regular tetrahedron $T \subset \mathbb{R}^3$ into n strictly congruent pieces is a classical problem in combinatorial geometry. Despite a century of piecewise progress—from Hill’s parallelepipeds [8] through the median-plane decompositions of Alexanderson and Wetzel [1], the algorithmic refinements of Ong [10], the longest-edge bisection of Padrón–Plaza–Suárez [11], the Dehn invariant technology following Hadwiger [7] and Conway–Jones [3], and the symmetry-ring language of Burnside [2, 13]—no unified treatment has appeared. In particular the following natural conjecture has remained open:

Conjecture 1.1 (Divisor Completeness). n admits a congruent dissection of the regular tetrahedron if and only if $n \mid 24$.

The purpose of the present paper is fourfold.

First, we provide a computational atlas of the eight historically known dissections $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ (Section 3), with canonical Cartesian coordinates, exact squared-edge spectra, and independently verified metric data. We establish in particular that *every* piece of each of the eight historical representatives is a tetrahedron (simplex with f -vector $(4, 6, 4)$) – in contrast to a persistent misconception in informal sources. The atlas covers the historical divisor representatives and does not aim at classifying arbitrary convex Coxeter-pure partitions;

see the companion work [4] for the convex Coxeter-pure catalogue, which includes pieces other than tetrahedra (e.g. an H-shape pentahedron at $n = 8$).

Second, we determine the refinement poset on the eight historical representatives (Section 5–Section 6), including the $S_4 \rightarrow S_8$ refinement bridging the Alexanderson–Wetzel and Padrón–Plaza–Suárez traditions, the disproof of the conjectured $S_6 \rightarrow S_{12}$ refinement, and the unique *heterogeneous* bridge $S_2 = S_3 + S_6$ at cut parameter $t = 2/3$.

Third, we develop three independent algebraic obstruction frameworks for conjecture 1.1: the Burnside ring $\text{Burn}(T_d)$ with its 11-class subgroup lattice (Section 9); the 24-cell and binary tetrahedral correspondence $2T \leftrightarrow T_d \leftrightarrow \text{flags}(T)$ (Section C); and a formal grammar of affine contractions encoding all halving refinements (Section 10). Together these realise every divisor $n \mid 24$ and, in the G_{diss} -transitive case, force $n \mid 24$ by Lagrange’s theorem applied to $G_{\text{diss}}(D) \leq T_d$ (Definition 2.4 and corollary 7.4). The remaining open gap is precisely:

Conjecture 1.2 (Transitivity Conjecture). Every congruent dissection of the regular tetrahedron is G_{diss} -transitive in the sense of Definition 2.4.

Relation to the Coxeter-pure classification. The present atlas is complementary to the companion paper [4], which classifies all convex Coxeter-pure (chamber-union) congruent partitions of T into 15 S_4 -orbit families. The two scopes are distinct: [4] fixes the A_3 Coxeter complex as the underlying combinatorics but allows arbitrary convex pieces (including non-tetrahedral ones, e.g. a pentahedral H-shape at $n = 8$), whereas the present paper fixes the eight historical divisor representatives and develops obstruction frameworks for non-Coxeter / non-transitive dissections. The naming crosswalk $T_i \leftrightarrow S_j$ of [4] is compatible with the atlas labels S_n used here; see Section 3.1.

Fourth, in Section 12 we attack the reverse direction via a structured programme whose two principal new results are:

Conditional Theorem 1.3 ($n = 5$ boundary-role obstruction, preview). *Under the explicit boundary-role homogeneity condition BRH, no partition of T into 5 full-dimensional congruent convex pieces exists. BRH requires one fixed set of reference facets to be exposed in every copy; it is not a consequence of congruence alone. The unrestricted mixed-role case remains open. See Conditional Theorem 12.16 and Section 12.5 for the precise statement and proof.*

The $n = 5$ contribution is therefore a conditional theorem plus exact finite bookkeeping, rather than an unconditional residual reduction. The corrected real-area enumeration and the local NN identity are included to make the remaining hypotheses visible; the historical phase-26 reduction scripts are retained as audit material but are not used as proof evidence.

Conditional Theorem 1.4 ($n = 16$ all-one-face, preview). *Under the medial-role and slot-separation hypotheses $(H_{\text{roles}}) + (H_{\text{slot}})$, no face-to-face convex congruent dissection of T into 16 pieces can place exactly one boundary face of every piece on a single common T -face (general convex instance: conditional theorem 12.8; Coxeter-pure instance: unconditional, theorem 12.7). See Section 12.4 for the full statement, the list of rigidity lemmas used, and the proof.*

Evidence levels. We adopt throughout the evidence-level tagging scheme L0 (heuristic), L1 (partial evidence), L1+ (strong partial evidence), L2 (conditional theorem), L2+ (finite-check conditional theorem), L3 (unconditional theorem); see Section B for the precise definitions. Every claim in this paper is tagged with one of these six levels, and nothing else: hybrid tags such as L2[−] or L1+/L2[−] are *not* used. The associated computational artifacts (source code, test suites, JSON certificates) are publicly available in the companion repository; see the *Data and code availability* statement at the end of Section 1.

Organisation. Section 3 presents the atlas; Section 4–Section 8 the Dehn invariant verification, the refinement poset, transitivity plus the Lagrange obstruction, and the comparison with

the cube. [Section 9](#)–[Section 10](#) develop the three algebraic obstruction frameworks (Burnside ring, IFS grammar; the 24-cell viewpoint is deferred to [Section C](#)); [Section 11](#) the non-transitive impossibility analysis (obstruction stack under T_d -invariance); [Section 12](#) the rigidity and configurational programme toward [conjecture 1.2](#), including [conditional theorems 1.3](#) and [1.4](#).

Acknowledgements. Computational certificates were produced with Python `Fractions` (exact rational arithmetic) and a custom MILP model built on PuLP/CBC.

Data and code availability. The manuscript source, test suites, scripts, and JSON certificates released with this paper are publicly available at <https://github.com/aconsciousfractal/Congruent-Dissections-of-the-Regular-Tetrahedron>. Executable artifacts are stored under `scripts/`, with deterministic outputs under `results/`; these include `results/S6_to_S12_edge_spectrum.json` and the active $n = 5$ BRH, area, and local-identity certificates. The general-convex $n = 16$ statement is explicitly conditional on $(H_{\text{roles}}) + (H_{\text{slot}})$; no separate public computational discharge of those two hypotheses is claimed. This release has no Zenodo DOI.

2 Preliminaries: four notions of symmetry

The tetrahedral group $T_d \cong S_4$ acts isometrically on T by permuting its four vertices; this lifts to a linear action on $\mathbb{R}^3$ fixing the centroid of T , and hence to an action on the power set of T sending convex sub-polyhedra to convex sub-polyhedra of the same congruence class. In the sequel we use four distinct notions of symmetry attached to a congruent partition $D = \{P_1, \dots, P_n\}$ of T ; they are logically independent and have been repeatedly conflated in earlier literature.

Definition 2.1 (Piece-type homogeneous space). Fix a piece $P_0 \in D$ and let $\text{Stab}_{T_d}(P_0) = \{g \in T_d : g(P_0) = P_0\}$ be its pointwise stabiliser under the T_d -action on convex sub-polyhedra of T . The *piece-type homogeneous space* of P_0 is the transitive T_d -set $T_d / \text{Stab}_{T_d}(P_0)$. Its cardinality equals the size of the T_d -orbit $T_d \cdot P_0 \subset 2^T$.

Definition 2.2 (Geometric symmetry group of a partition). The *geometric symmetry group* of D is

$$G_{\text{diss}}(D) := \{g \in T_d : g(D) = D \text{ as a set of pieces}\} \leq T_d,$$

i.e. the subgroup of T_d that permutes the pieces of D among themselves.

Definition 2.3 (T_d -invariance). D is *T_d -invariant* if $G_{\text{diss}}(D) = T_d$, i.e. every $g \in T_d$ permutes the pieces of D .

Definition 2.4 (G_{diss} -transitivity). D is *G_{diss} -transitive* if $G_{\text{diss}}(D)$ (acting by its inclusion $G_{\text{diss}}(D) \leq T_d$) acts transitively on the set of pieces of D : for every $P_i, P_j \in D$ there exists $g \in G_{\text{diss}}(D)$ with $g(P_i) = P_j$. Equivalently, the n pieces form a single orbit under the group of T_d -elements that preserve D set-wise.

Remark 2.5 (Logical relations). The four notions are related as follows.

1. T_d -invariance ([Definition 2.3](#)) is the strongest condition and implies $G_{\text{diss}}(D) = T_d$; in the atlas it holds only for $n \in \{1, 12, 24\}$ ([Lemma 9.2](#)).
2. G_{diss} -transitivity ([Definition 2.4](#)) is the notion that yields the Lagrange obstruction $n \mid 24$ ([Corollary 7.4](#)): orbit–stabiliser in $G_{\text{diss}}(D)$ gives $n = [G_{\text{diss}}(D) : \text{Stab}_0]$, and $|G_{\text{diss}}(D)|$ divides $|T_d| = 24$ by Lagrange applied to $G_{\text{diss}}(D) \leq T_d$.
3. T_d -invariance $\Rightarrow$ G_{diss} -transitivity (whenever T_d itself acts transitively on D); the converse fails for $n \in \{2, 3, 4, 6, 8\}$, where $G_{\text{diss}}(S_n) \subsetneq T_d$.

4. The *piece-type* homogeneous space $T_d/\text{Stab}_{T_d}(P_0)$ of Definition 2.1 can have more elements than the partition D : an element $g \in T_d \setminus G_{\text{diss}}(D)$ maps P_0 to a convex sub-polyhedron of T that lies in a *different* congruent partition. Consequently T_d -type properties (e.g. the order $24/n$ quoted in Table 4) are formal identities derived from $|T_d| = 24$ and $|D| = n$, not direct statements about pointwise stabilisers of specific realisations.

Whenever the word *transitive* appears unqualified in the sequel, it refers to Definition 2.4.

3 The computational atlas $\mathcal{F}(T)$

Through exact symbolic computation we map each of the eight known dissections into canonical Cartesian coordinates with full metric verification: n congruent pieces, each of volume $\text{Vol}(T)/n$, tiling T without gap or overlap. Throughout this section the tetrahedron T is embedded with vertices

$$A = (0, 0, 0), B = (1, 0, 0), C = (\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}, 0), D = (\tfrac{1}{2}, \tfrac{\sqrt{3}}{6}, \tfrac{\sqrt{6}}{3}), \quad (1)$$

so that $|AB| = |AC| = \dots = |CD| = 1$. We write M_{XY} for the midpoint of edge $\overline{XY}$, G_{XYZ} for the centroid of face $\triangle XYZ$, and G_T for the centroid of T .

3.1 Historical atlas

Table 1: The eight known congruent dissections of T . Each piece has f -vector $(4, 6, 4)$: all eight piece types are tetrahedra. Squared edge spectrum E^2 is normalised to the parent edge length $|AB| = 1$.

n	Piece name	f -vector	E^2 (squared edges)	Construction
1	Regular tetrahedron	$(4, 6, 4)$	$[1, 1, 1, 1, 1, 1]$	Identity
2	Median wedge	$(4, 6, 4)$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}, 1, 1, 1]$	1 median plane
3	Axial wedge	$(4, 6, 4)$	$[\frac{1}{3}, \frac{1}{3}, \frac{2}{3}, 1, 1, 1]$	3 axial planes
4	Quadrisection	$(4, 6, 4)$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, 1]$	2 median planes
6	Hexasection	$(4, 6, 4)$	$[\frac{1}{12}, \frac{1}{4}, \frac{1}{3}, \frac{2}{3}, \frac{3}{4}, 1]$	3 axial + 3 median
8	8T-LE piece	$(4, 6, 4)$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}]$	longest-edge iteration
12	A_4 domain	$(4, 6, 4)$	$[\frac{1}{24}, \frac{1}{3}, \frac{1}{3}, \frac{3}{8}, \frac{3}{8}, 1]$	A_4 fundamental domain
24	Orthoscheme	$(4, 6, 4)$	$[\frac{1}{24}, \frac{1}{12}, \frac{1}{8}, \frac{1}{4}, \frac{1}{3}, \frac{3}{8}]$	6 median planes

Remark on the $n = 12$ entry. By “ A_4 fundamental domain” we mean a fundamental domain for the action of the alternating group A_4 , the orientation-preserving subgroup of $T_d \cong S_4$, on T : any such domain is a 12-piece dissection piece, concretely realised as $\text{conv}(A, G_{ABC}, G_T, B)$ (Table 2), i.e. the convex hull of a vertex, a face barycentre, the T -centroid, and an edge-adjacent vertex.

Table 2: Canonical representative pieces in the embedding (1).

n	Representative S_n
2	$\text{conv}(A, B, C, M_{CD})$
3	$\text{conv}(D, G_{ABC}, A, B)$
4	$\text{conv}(A, M_{AB}, C, M_{CD})$
6	$\text{conv}(D, G_{ABC}, A, M_{AB})$
8	$\text{conv}(A, M_{AB}, M_{AC}, M_{CD})$
12	$\text{conv}(A, G_{ABC}, G_T, B)$
24	$\text{conv}(A, M_{AB}, G_{ABC}, G_T)$

Theorem 3.1 (Tetrahedrality of the eight historical representatives). *Every piece of each of the eight historical congruent dissection representatives $\{S_n\}_{n \in \text{div}(24)}$ listed in Table 2 is a tetrahedron.*

Proof. Direct verification from Table 2: each representative is the convex hull of exactly four affinely independent points. The informal description of S_3 as a “wedge” pentahedron in certain forum sources is a notational artefact and not the actual piece geometry; similarly, the S_{12} piece obtained by merging two adjacent orthoschemes reduces to a tetrahedron because the merged midpoint M_{AB} lies on edge $\overline{AB}$ and drops out of the convex hull. $\square$

Remark 3.2 (Scope). theorem 3.1 is a statement about the eight specific historical representatives, not about all Coxeter-pure congruent partitions of T . The companion classification [4] exhibits convex Coxeter-pure pieces which are not tetrahedra; in particular at $n = 8$ the H-shape family $\mathcal{F}_{11}$ has f -vector $(6, 9, 5)$ and is a pentahedron. The present atlas fixes the eight classical representatives; the general convex Coxeter-pure case is addressed in [4].

4 Dehn invariants of the atlas pieces

The Dehn invariant $D(P) \in \mathbb{R} \otimes_{\mathbb{Z}} (\mathbb{R}/\pi\mathbb{Q})$ is additive under scissors decomposition (Sydler [12]; see also Dupont [6] for the modern group-homological formulation), and the regular tetrahedron carries the irrational value $D(T) = 6 \otimes \alpha$ where $\alpha := \arccos(1/3) \approx 70.529^\circ$ is the dihedral angle along every edge. By additivity a congruent n -dissection $T = \bigcup S_n$ requires

$$D(S_n) = \frac{1}{n} D(T) = \frac{6}{n} \otimes \alpha. \quad (2)$$

Computation of the piece parameters reveals a conjugate dihedral angle $\beta := \arccos(\sqrt{6}/3)$. By the trigonometric double-angle identity one finds $\cos(2\beta) = 2 \cdot (6/9) - 1 = 1/3$. Since both 2β and α lie in $(0, \pi)$, this gives the exact identity

$$2\beta = \alpha. \quad (3)$$

This local angle identity is not a classification of all individual dihedral angles of the atlas pieces. The total Dehn tensors below follow from additivity and congruence, independently of such a classification.

Corollary 4.1 (Dehn values of the atlas pieces). *Applying (2) to the eight congruent atlas dissections gives $D(S_1) = 6 \otimes \alpha$ and the following values:*

n	2	3	4	6	8	12	n	24
$D(S_n)$	$3 \otimes \alpha$	$2 \otimes \alpha$	$\frac{3}{2} \otimes \alpha$	$1 \otimes \alpha$	$\frac{3}{4} \otimes \alpha$	$\frac{1}{2} \otimes \alpha$	$D(S_n)$	$\frac{1}{4} \otimes \alpha$

Corollary 4.2 (No cube-equivalent piece). *No piece S_n of any known congruent dissection of T has $D(S_n) = 0$; consequently no such piece is scissors-congruent to a cube.*

5 The structural lineage $S_4 \rightarrow S_8$

The $8T$ -LE construction of Padrón, Plaza and Suárez [11] is visually unlike the global median-plane decompositions of Alexanderson–Wetzel [1]: it is generated by iterated *longest-edge bisection*, a local algorithmic procedure. Canonical coordinates however reveal a clean lineage.

Observation 5.1 (Canonical lineage). The $8T$ -LE dissection descends from the S_4 quadrisection as follows.

1. Bisection of the longest edge of T (any edge, by symmetry) yields S_2 .
2. Bisection of the longest edge of S_2 introduces a strictly orthogonal plane (perpendicular to the first median plane), yielding S_4 .

3. The longest edge of S_4 is the unique remaining original T -edge of length 1; bisecting it divides S_4 into two copies of S_8 .

The refinement graph over halving cuts is summarised in Figure 1. The axial and symmetry-plane pipelines are geometrically *incompatible* at the $S_6 \rightarrow S_{12}$ junction: the S_6 representative $\text{conv}(D, G_{ABC}, A, M_{AB})$ uses the apex D , while $S_{12} = \text{conv}(A, G_{ABC}, G_T, B)$ uses the tetrahedron centroid G_T and a fresh vertex B . The two families thus form disjoint components connected only through the heterogeneous bridge $S_2 = S_3 + S_6$ described in Section 6.

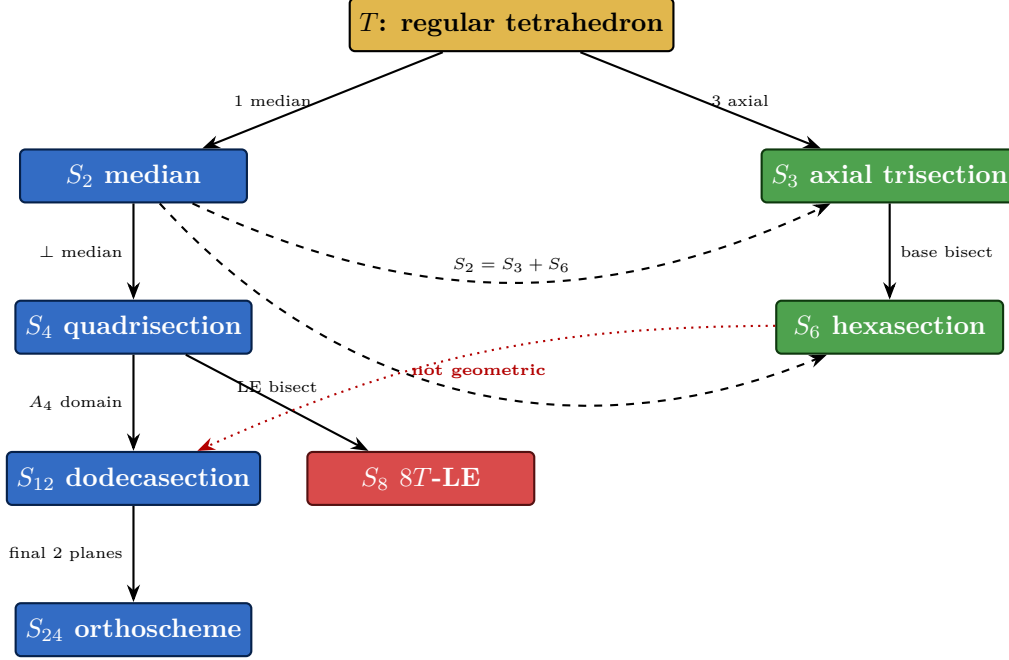


Figure 1: The refinement poset of the eight known dissections. Solid arrows are verified halving refinements; dashed arrows record the unique heterogeneous bridge $S_2 = S_3 + S_6$ (at cut parameter $t = 2/3$, see Section 6); the dotted red arrow marks the geometrically *disproved* refinement $S_6 \rightarrow S_{12}$. Blue = symmetry pipeline, green = axial pipeline, red = longest-edge pipeline.

Theorem 5.2 (Disproof of $S_6 \rightarrow S_{12}$). *There is no congruent halving of the S_6 piece into two tetrahedra each congruent to S_{12} .*

Proof. The argument is by the squared-edge multiset, a congruence invariant: if two tetrahedra P, Q are congruent then the unordered multisets $E^2(P), E^2(Q)$ of squared edge lengths coincide, and in particular $\min E^2(P) = \min E^2(Q)$. From Table 1 we read

$$E^2(S_{12}) = \left\{ \frac{1}{24}, \frac{1}{3}, \frac{1}{3}, \frac{3}{8}, \frac{3}{8}, 1 \right\}, \quad \min E^2(S_{12}) = \frac{1}{24}. \quad (4)$$

Shape of a congruent halving. A plane cut of a tetrahedron P produces two tetrahedra if and only if the cut plane contains exactly one edge of P and meets the opposite edge at a single interior point (any cut meeting three edges produces a tetrahedron and a triangular prism, hence cannot yield two tetrahedra). If in addition the two halves are congruent – and hence of equal volume $\frac{1}{2} \text{Vol}(P)$ – the interior intersection point must be the midpoint of the opposite edge. Applying this to $S_6 = \text{conv}(D, G_{ABC}, A, M_{AB})$ leaves exactly three opposite-edge pairs and hence six candidate half-tetrahedra.

Exact enumeration in the embedding (1). A direct computation in rational arithmetic (certificate `results/S6_to_S12_edge_spectrum.json` in the public companion repository; see the *Data and code availability* statement in Section 1) gives the following squared-edge multisets for the

six candidate halves, indexed by the edge e preserved by the cut plane and the vertex v of $\{D, G_{ABC}, A, M_{AB}\} \setminus e$ retained in that half:

$$\begin{aligned} e = \{D, G_{ABC}\} : & \left\{ \frac{1}{16}, \frac{7}{48}, \frac{1}{3}, \frac{2}{3}, \frac{13}{16}, 1 \right\}, \left\{ \frac{1}{16}, \frac{1}{12}, \frac{7}{48}, \frac{2}{3}, \frac{3}{4}, \frac{13}{16} \right\}; \\ e = \{D, A\} : & \left\{ \frac{1}{48}, \frac{13}{48}, \frac{1}{3}, \frac{2}{3}, \frac{11}{16}, 1 \right\}, \left\{ \frac{1}{48}, \frac{1}{4}, \frac{13}{48}, \frac{11}{16}, \frac{3}{4}, 1 \right\}; \\ e = \{D, M_{AB}\} : & \left\{ \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{2}{3}, \frac{3}{4}, \frac{3}{4} \right\}, \left\{ \frac{1}{12}, \frac{1}{12}, \frac{1}{4}, \frac{3}{4}, \frac{3}{4}, 1 \right\}. \end{aligned}$$

The set of minima attained across the six halves is $\{\frac{1}{48}, \frac{1}{16}, \frac{1}{12}\}$. By (4), $\min E^2(S_{12}) = \frac{1}{24}$ does not belong to this set, so no half has S_{12} 's squared-edge multiset. Hence no congruent halving of S_6 produces two copies of S_{12} . $\square$

6 The refinement graph

6.1 Egyptian fractions over $\text{div}(24)$

We exhaustively test single-plane congruent cuts $S_a \rightarrow S_b \cup S_c$ with $a, b, c \in \text{div}(24)$, $b \leq c$, $b > a$ and volume-consistent Egyptian fraction $1/a = 1/b + 1/c$; the target is geometric realisability (edge-spectrum congruence of the two halves).

Table 3: Egyptian fraction audit. Six of the eleven admissible identities are geometrically realisable.

Identity	Geometric?	Note
$1 = \frac{1}{2} + \frac{1}{2}$	✓	Median bisection
$\frac{1}{2} = \frac{1}{3} + \frac{1}{6}$	✓ (new)	Cross-pipeline bridge
$\frac{1}{2} = \frac{1}{4} + \frac{1}{4}$	✓	Orthogonal median
$\frac{3}{4} = \frac{1}{4} + \frac{1}{2}$	×	
$\frac{3}{4} = \frac{1}{6} + \frac{1}{6}$	✓	Base median bisection
$\frac{4}{4} = \frac{1}{6} + \frac{1}{2}$	×	
$\frac{4}{4} = \frac{1}{8} + \frac{1}{8}$	✓	8T-LE bisection
$\frac{6}{6} = \frac{1}{8} + \frac{1}{24}$	×	
$\frac{6}{6} = \frac{1}{12} + \frac{1}{12}$	×	$S_6 \rightarrow S_{12}$ disproved
$\frac{8}{8} = \frac{1}{12} + \frac{1}{24}$	×	volumes match, shapes do not
$\frac{1}{12} = \frac{1}{24} + \frac{1}{24}$	✓	orthoscheme split

6.2 The heterogeneous bridge $S_2 = S_3 + S_6$

Theorem 6.1 (Cross-pipeline bridge). *The single-plane cut of the S_2 representative $S_2 = \text{conv}(A, B, C, M_{CD})$ at cut parameter $t = 2/3$ along a cut-edge produces one piece with normalised squared edges $(1, 1, 2, 3, 3, 3)$ (matching S_3) and one piece with normalised squared edges $(1, 3, 4, 8, 9, 12)$ (matching S_6). This is the unique heterogeneous refinement bridge in the catalogue.*

The two concrete realisations are related by swapping the fixed edge; in both, $\det(\alpha) \cdot \det(\beta) = -1$, so one refinement map is orientation-preserving and one is orientation-reversing.

6.3 Connected components

Proposition 6.2 (Component structure). *Under homogeneous (1:1) halving refinements the refinement graph has three disconnected components:*

Component	Nodes	Pipeline
1	$\{S_1, S_2, S_4, S_8\}$	Symmetry + LE
2	$\{S_3, S_6\}$	Axial
3	$\{S_{12}, S_{24}\}$	A_4 / Orthoscheme

The heterogeneous bridge of [theorem 6.1](#) connects components 1 and 2; components 2 and 3 are geometrically disjoint.

7 T_d -transitivity and the Lagrange obstruction

7.1 All eight known dissections are G_{diss} -transitive

Computational Certificate 7.1 (G_{diss} -transitivity of the atlas). *For every $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ the atlas dissection S_n is G_{diss} -transitive in the sense of [Definition 2.4](#): the geometric symmetry group $G_{\text{diss}}(S_n) \leq T_d$ of [Definition 2.2](#) acts transitively on the n pieces of S_n , with orbit-stabiliser identity $|G_{\text{diss}}(S_n)| = n \cdot |\text{Stab}_0|$ ([Lemma 9.2](#)). The numerical data of this certificate are tabulated in [Tables 4](#) and [10](#).*

Remark 7.2 (Reading [Table 4](#)). The column $|\text{Stab}|$ of [Table 4](#) records the formal identity $|\text{Stab}| = 24/n$; for $n \in \{1, 12, 24\}$ it coincides with the pointwise stabiliser of a piece in T_d (since $G_{\text{diss}} = T_d$), while for $n \in \{2, 3, 4, 6, 8\}$ the pointwise T_d -stabiliser of a specific realisation is strictly smaller and the correct orbit-stabiliser decomposition takes place inside $G_{\text{diss}}(S_n) \subsetneq T_d$ ([Lemma 9.2](#)). The value $24/n$ nonetheless correctly controls the Lagrange obstruction via $|G_{\text{diss}}| \mid 24$.

Table 4: The T_d -action on each of the eight known dissections.

n	Orbit size	$ \text{Stab} $	Stabiliser type
1	1	24	T_d
2	2	12	A_4
3	3	8	D_{2d}
4	4	6	S_3
6	6	4	V_4 (Klein four-group)
8	8	3	$\mathbb{Z}_3$
12	12	2	$\mathbb{Z}_2$
24	24	1	$\{e\}$

Remark 7.3 (On the $n = 8$ anomaly). Although the 8T-LE dissection is generated by an iterative algorithmic procedure rather than as a subgroup orbit, T_d nevertheless acts transitively on its 8 pieces, with stabiliser $\mathbb{Z}_3$ generated by the 120° rotation about a body diagonal. In particular $8 \mid 24$ is *structural*, not accidental.

7.2 The Lagrange obstruction

The immediate consequence of [computational certificate 7.1](#) is:

Corollary 7.4 (Divisor realisation and Lagrange obstruction). *If $n \mid 24$, the regular tetrahedron admits a congruent dissection into n pieces. Moreover, for any G_{diss} -transitive congruent dissection of T ([Definition 2.4](#)) necessarily $n \mid 24$.*

Both statements follow by construction ([Table 1](#)) and by applying Lagrange’s theorem to the chain $\text{Stab}_0 \leq G_{\text{diss}}(D) \leq T_d$: $n = [G_{\text{diss}}(D) : \text{Stab}_0]$ divides $|G_{\text{diss}}(D)|$, which in turn divides $|T_d| = 24$. The full divisor-completeness conjecture thus reduces to ruling out congruent dissections that fail G_{diss} -transitivity – the subject of [Sections 11](#) and [12](#).

8 Comparison with the cube

To probe whether [conjecture 1.1](#) generalises to other Platonic solids we analyse the cube (symmetry group O_h , $|O_h| = 48$).

Proposition 8.1 (Slab argument). *The unit cube $[0, 1]^3$ admits a congruent dissection into n pieces for every $n \geq 1$: cut by planes $z = k/n$, $k = 1, \dots, n-1$, to produce n congruent rectangular slabs $1 \times 1 \times (1/n)$, all related by translation.*

The mechanism of [proposition 8.1](#) requires *parallel faces* on the parent body. Among all Platonic solids the tetrahedron is the unique one with *zero* parallel face-pairs ([Table 5](#)), so the slab argument cannot be transplanted to it.

Table 5: Parallel-face count and slab cuts among the Platonic solids.

Solid	Group	$ G $	Parallel face-pairs	Slab cuts?
Tetrahedron	T_d	24	0	No
Cube	O_h	48	3	Yes
Octahedron	O_h	48	4	Yes
Dodecahedron	I_h	120	6	Yes
Icosahedron	I_h	120	6	Yes

A refined question remains valid on the cube: for which n does the cube admit an O_h -*transitive* congruent dissection? The answer is exactly $n \mid 48$, with explicit O_h -transitive constructions for all ten divisors of 48 ([Table 6](#)).

Table 6: O_h -transitive congruent dissections of the cube.

n	$ \text{Stab} $	Construction
1	48	Trivial
2	24	Mirror-plane bisection
3	16	Double-wedge (bicapped square)
4	12	Two orthogonal mirror planes
6	8	Six face pyramids
8	6	$2 \times 2 \times 2$ cubelets
12	4	Bisection of six pyramids
16	3	Grid subdivisions
24	2	O -orbit orthoscheme
48	1	O_h -orbit orthoscheme

Corollary 8.2 (Failure of divisor completeness for the cube). *Divisor completeness does not hold for the cube: all $n \geq 1$ admit non-transitive slab dissections. The emerging dichotomy is:*

	Tetrahedron (T_d)	Cube (O_h)
<i>G-transitive: $n \mid G$ sufficient?</i>	✓ (8 divisors)	✓ (10 divisors)
<i>Non-G-transitive constructions</i>	none known	all n (slabs)
<i>Parallel faces</i>	0	3

Hence [conjecture 1.1](#) is *not* a generic polyhedral phenomenon: it is specifically the absence of parallel faces on the regular tetrahedron that prevents non-transitive constructions and motivates [conjecture 1.2](#).

Remark 8.3 (Arithmetic note). The aggregate divisor sums do not transfer:

$$\frac{\sigma(24)}{24} = \frac{60}{24} = \frac{5}{2}, \quad \frac{\sigma(48)}{48} = \frac{124}{48} = \frac{31}{12}.$$

9 The Burnside ring invariant

The Burnside ring $\text{Burn}(T_d)$ [[2](#), [13](#)] provides a representation-theoretic framework for characterising transitive G -sets and detecting obstructions to non-transitive dissections.

9.1 The subgroup lattice of $T_d \cong S_4$

The tetrahedral group $T_d \cong S_4$ has 11 *conjugacy classes of subgroups* and 30 subgroups in total (Table 7).

Table 7: The 11 conjugacy classes of subgroups of S_4 .

Class	Subgroup H	$ H $	$[S_4 : H]$	# conjugates
0	$\{e\}$	1	24	1
1	$\mathbb{Z}_2^t$ (transposition)	2	12	6
2	$\mathbb{Z}_2^d$ (double transposition)	2	12	3
3	$\mathbb{Z}_3$	3	8	4
4	V_4 (non-normal Klein-4)	4	6	3
5	V_4^n (normal Klein-4)	4	6	1
6	$\mathbb{Z}_4$	4	6	3
7	S_3	6	4	4
8	D_4	8	3	3
9	A_4	12	2	1
10	S_4	24	1	1

The valid orbit sizes for any T_d -action are thus $\{1, 2, 3, 4, 6, 8, 12, 24\}$ – precisely the eight divisors of 24.

9.2 Equivariant Euler characteristic

The barycentric subdivision of T yields a T_d -equivariant CW-complex. A direct count gives

$$\chi(T) = 15 - 50 + 60 - 24 = 1, \quad (5)$$

consistent with T being a contractible 3-ball. Lifting to the Burnside ring:

Computational Certificate 9.1 (Equivariant Euler class). *The equivariant Euler characteristic $\chi_{T_d}(T)$ of T , viewed as an equivariant CW-complex, equals the class of the point: $\chi_{T_d}(T) = [T_d/T_d] \in \text{Burn}(T_d)$, with character vector $(1, 1, \dots, 1) \in \mathbb{Z}^{11}$.*

9.3 The mark table

The mark table $M[i][j] = |\text{Fix}_{H_j}(T_d/H_i)|$ encodes the fixed-point structure of every transitive T_d -set. Its computed value is given in Table 8; it has full rank 11, confirming that $\text{Burn}(T_d)$ is the free abelian group of rank 11 on the transitive T_d -sets.

Table 8: Mark table of $T_d \cong S_4$. Rows indexed by transitive T_d -sets T_d/H_i (top-to-bottom: subgroup order ascending); columns indexed by subgroups H_j (left-to-right: same ordering).

T_d/H_i	$\{e\}$	$\mathbb{Z}_2^t$	$\mathbb{Z}_2^d$	$\mathbb{Z}_3$	V_4	V_4^n	$\mathbb{Z}_4$	S_3	D_4	A_4	S_4
$T_d/\{e\}$	24	0	0	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_2^t$	12	2	0	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_2^d$	12	0	4	0	0	0	0	0	0	0	0
$T_d/\mathbb{Z}_3$	8	0	0	2	0	0	0	0	0	0	0
T_d/V_4	6	2	2	0	2	0	0	0	0	0	0
T_d/V_4^n	6	0	6	0	0	6	0	0	0	0	0
$T_d/\mathbb{Z}_4$	6	0	2	0	0	0	2	0	0	0	0
T_d/S_3	4	2	0	1	0	0	0	1	0	0	0
T_d/D_4	3	1	3	0	1	3	1	0	1	0	0
T_d/A_4	2	0	2	2	0	2	0	0	0	2	0
T_d/S_4	1	1	1	1	1	1	1	1	1	1	1

9.4 Coset decomposition – algebra matches geometry

For every known dissection the Burnside ring element is a single transitive T_d -set:

Table 9: Each dissection is a single transitive T_d -set in $\text{Burn}(T_d)$.

S_n	Coset space	Stabiliser	$\text{Burn}(T_d)$ -element
S_1	T_d/S_4	S_4	$[T_d/S_4]$
S_2	T_d/A_4	A_4	$[T_d/A_4]$
S_3	T_d/D_4	D_4	$[T_d/D_4]$
S_4	T_d/S_3	S_3	$[T_d/S_3]$
S_6	T_d/V_4	V_4	$[T_d/V_4]$
S_8	$T_d/\mathbb{Z}_3$	$\mathbb{Z}_3$	$[T_d/\mathbb{Z}_3]$
S_{12}	$T_d/\mathbb{Z}_2^t$	$\mathbb{Z}_2^t$	$[T_d/\mathbb{Z}_2^t]$
S_{24}	$T_d/\{e\}$	$\{e\}$	$[T_d/\{e\}]$

9.5 Geometric vs. algebraic stabilisers

We now make precise the relation between the two notions of symmetry attached to the atlas: the piece-type stabiliser of [Definition 2.1](#) (recorded in [Table 4](#), “algebraic”) and the partition-level stabiliser G_{diss} of [Definition 2.2](#) (recorded in [Table 10](#), “geometric”).

Lemma 9.2 (Geometric vs. algebraic stabilisers). *For each S_n in the atlas let P_0 be the canonical piece of [Table 2](#), let $H_n := \text{Stab}_{T_d}(P_0)$ be its piece-type stabiliser ([Definition 2.1](#), $|H_n| = 24/n$), and let $G_{\text{diss}}(S_n)$ be its geometric symmetry group ([Definition 2.2](#)). Then:*

1. $G_{\text{diss}}(S_n)$ is the subgroup of T_d listed in [Table 10](#), and the orbit–stabiliser identity $|G_{\text{diss}}(S_n)| = n \cdot |\text{Stab}_0|$ holds for every n , where $\text{Stab}_0 = G_{\text{diss}}(S_n) \cap H_n$ is the stabiliser of P_0 inside the geometric group.
2. $G_{\text{diss}}(S_n) = T_d$ (i.e. S_n is T_d -invariant in the sense of [Definition 2.3](#)) for $n \in \{1, 12, 24\}$, and only for these n .
3. For $n \in \{2, 3, 4, 6, 8\}$ one has $G_{\text{diss}}(S_n) \subsetneq T_d$; the elements of $T_d \setminus G_{\text{diss}}(S_n)$ send S_n to a distinct but congruent partition of T .
4. In every case S_n is G_{diss} -transitive ([Definition 2.4](#)), i.e. the n pieces form a single orbit under $G_{\text{diss}}(S_n)$; this is the statement certified in [Computational Certificate 7.1](#).

Proof. The numerical data of [Table 10](#) are verified by direct computation in the embedding (1): for each S_n we enumerate the 24 elements $g \in T_d$ and test whether the image partition $g(S_n)$ equals S_n setwise; the set of such g is $G_{\text{diss}}(S_n)$. Items (2) and (3) are read off the table. The orbit–stabiliser identity in (1) is the usual one for the transitive action of $G_{\text{diss}}(S_n)$ on the n pieces of S_n ; transitivity within $G_{\text{diss}}(S_n)$ is verified case by case by the atlas computation. The public script `scripts/independent_verifier.py` gives a separate check in the stated Coxeter-pure 15-family scope. Item (4) is a restatement of the definitions once transitivity of the G_{diss} -action is established. $\square$

Remark 9.3 (Summary). For $n \in \{1, 12, 24\}$ all four notions of [Section 2](#) coincide; for $n \in \{2, 3, 4, 6, 8\}$ only G_{diss} -transitivity ([Definition 2.4](#)) holds, and $G_{\text{diss}}(S_n) \subsetneq T_d$ is strictly smaller than T_d . In particular the Lagrange obstruction $n \mid 24$ ([Corollary 7.4](#)) uses only G_{diss} -transitivity.

Table 10: Geometric vs. algebraic stabilisers of the eight dissections.

S_n	G_{diss}	$ G_{\text{diss}} $	Transitive?	$ \text{Stab}_0 $
S_1	S_4	24	✓	24
S_2	V_4	4	✓	2
S_3	S_3	6	✓	2
S_4	D_4	8	✓	2
S_6	S_3	6	✓	1
S_8	D_4	8	✓	1
S_{12}	S_4	24	✓	2
S_{24}	S_4	24	✓	1

10 IFS dissection grammar

Every halving refinement in the poset of [Sections 5](#) and [6](#) can be encoded as an iterated function system (IFS): a pair of affine contractions whose images tile the parent piece. This produces a formal grammar for generating all eight dissections.

10.1 Halving rules

Table 11: Halving rules and their chirality. Rule R3 ($S_4 \rightarrow S_8$) is the unique rule with *both* branches orientation-preserving.

Rule	Production	Geometric name	$\det(f_1)$	$\det(f_2)$
R1	$S_1 \rightarrow f_1(S_2) \cup f_2(S_2)$	Median bisection	+1	-1
R2	$S_2 \rightarrow g_1(S_4) \cup g_2(S_4)$	Orthogonal median	+1	-1
R3	$S_4 \rightarrow h_1(S_8) \cup h_2(S_8)$	Longest-edge bisection	+1	+1
R4	$S_3 \rightarrow p_1(S_6) \cup p_2(S_6)$	Axial median bisection	+1	-1
R5	$S_{12} \rightarrow q_1(S_{24}) \cup q_2(S_{24})$	Edge bisection	+1	-1

10.2 Heterogeneous bridge (R6)

The heterogeneous bridge ([theorem 6.1](#)) corresponds to the rule $R6: S_2 \rightarrow \alpha(S_3) \cup \beta(S_6)$ with $\det(\alpha)\det(\beta) = -1$.

10.3 Formal alphabet and grammar

The alphabet of the grammar is $\Sigma = \{S_1, S_2, S_3, S_4, S_6, S_8, S_{12}, S_{24}\}$. Homogeneous productions are R1–R5; the unique heterogeneous production is R6. Derived multi-level compositions include $R1 \circ R2$ ($S_1 \rightarrow 4 \times S_4$, quadrisection) and $R1 \circ R2 \circ R3$ ($S_1 \rightarrow 8 \times S_8$, $8T$ -LE octasection).

10.4 Composition $S_1 \rightarrow 8 \cdot S_8$ and PPS23 similarity classes

Iterated longest-edge bisection of T produces a finite sequence of similarity classes [\[11\]](#); our grammar reproduces and extends their enumeration:

Table 12: Similarity class count under iterated longest-edge bisection.

Depth	Pieces	Similarity classes
1	2	1
2	4	2
3	8	4
4	16	8
5	32	9
6	64	12
7	128	16
8	256	17
9	512	18

The class count stabilises at 18 by depth 9, consistent with a finite attractor in similarity space. At depth 9 the four dominant classes are the regular tetrahedron ($\times 40$), an isosceles variant ($\times 40$), the S_8 -type ($\times 108$) and the S_3 -type ($\times 40$).

11 An obstruction stack for T_d -invariant dissections (screening)

Status of this section (honesty block). This section does *not* resolve the non- G_{diss} -transitive case of [conjecture 1.1](#). It builds an *evidence stack* that, under the standing hypothesis of T_d -invariance ([Definition 2.3](#)) and a T_d -maximality argument flagged as *plausible but not fully rigorous* ([Section 11.3](#) and [remark 11.4](#)), filters every tested orbit decomposition for $n \leq 16$, $n \nmid 24$. Of the nine such values of n four pass through the crystallographic restriction alone (*unconditional* screening in the T_d -invariant category) and five additionally invoke the T_d -maximality step, tagged ($\dagger$) in [Table 13](#) and recorded at level L1+ in [Table 14](#). A theorem-level resolution of the non-transitive problem, without the T_d -maximality caveat, is the content of the Stabiliser Transport Lemma ([Lemma 12.2](#), [Section 12](#)) and is not achieved here.

The remaining question for [conjecture 1.1](#) is whether a congruent n -dissection of T with $n \nmid 24$ could exist without being G_{diss} -transitive ([Definition 2.4](#)). We develop a systematic obstruction framework combining the subgroup lattice, the crystallographic restriction theorem, and a T_d -maximality argument; the resulting stack filters all tested decompositions in the T_d -invariant category ([Definition 2.3](#)). It is important to stress that T_d -invariance is strictly stronger than mere non-transitivity under G_{diss} : a hypothetical congruent dissection might have $G_{\text{diss}}(D) \subsetneq T_d$ and therefore escape the T_d -orbit analysis of this section entirely.

11.1 Orbit decomposition

Suppose a congruent n -dissection $T = \bigsqcup_{i=1}^n P_i$ is T_d -invariant ([Definition 2.3](#)), i.e. every $\sigma \in T_d$ permutes the pieces (equivalently, $\sigma(\{P_1, \dots, P_n\}) = \{P_1, \dots, P_n\}$ for all σ). All eight known dissections enjoy this property for $n \in \{1, 12, 24\}$ ([lemma 9.2](#)); for $n \in \{2, 3, 4, 6, 8\}$ only the smaller group $G_{\text{diss}}(S_n) \subsetneq T_d$ preserves the partition set-wise and the analysis below strictly speaking applies only after replacing T_d by $G_{\text{diss}}(S_n)$. The present section enforces T_d -invariance as a standing hypothesis so as to have the full group T_d available for the orbit decomposition. Under this assumption, the T_d -action decomposes into orbits $\mathcal{O}_1, \dots, \mathcal{O}_k$ with $|\mathcal{O}_j| = [T_d : H_j]$ for stabiliser $H_j \leq T_d$. Since each orbit size divides $|T_d| = 24$:

$$n = \sum_{j=1}^k [T_d : H_j], \quad [T_d : H_j] \in \{1, 2, 3, 4, 6, 8, 12, 24\}. \quad (6)$$

For $n \nmid 24$ necessarily $k \geq 2$.

Lemma 11.1 (Size-1 orbit). *At most one orbit has size 1.*

Proof. A T_d -invariant piece must contain the centroid G_T , the unique T_d -fixed point of T . Two distinct convex pieces containing G_T would overlap. $\square$

11.2 Cross-orbit symmetry obstruction

Let $P_1 \in \mathcal{O}_1$ and $P_2 \in \mathcal{O}_2$ be pieces in distinct orbits of a congruent T_d -invariant dissection. Since $P_1 \cong P_2$, a fixed congruence identifies them with a common abstract reference tile P whose intrinsic symmetry group is $K := \text{Sym}(P) \leq O(3)$. The stabilisers

$$H_1 = \text{Stab}_{T_d}(P_1), \quad H_2 = \text{Stab}_{T_d}(P_2)$$

act on P and hence embed (as subgroups, up to conjugation in $O(3)$) into K .

Lemma 11.2 (Stabiliser pair screening). *Let $H_1, H_2 \leq T_d$ be two point-stabiliser conjugacy classes from the 30 subgroups of T_d (in 11 conjugacy classes). Let $K(H_1, H_2) \leq O(3)$ denote the smallest finite subgroup of $O(3)$, up to conjugation, that contains a pair of non-conjugate copies of H_1 and H_2 (whenever such a pair exists). Then, using the classical classification of finite subgroups of $O(3)$ and the crystallographic restriction theorem [9], the following non-conjugate stabiliser pairs force K to contain a full cubic group $\geq S_4$:*

Non-conjugate pair (H_1, H_2)	Minimal containing $K \leq O(3)$
$(D_4, \mathbb{Z}_3)$	$\supseteq S_4$ (mixed 4-fold & 3-fold)
$(\mathbb{Z}_4, \mathbb{Z}_3)$	$\supseteq S_4$ (same mechanism)
(S_3, V_4)	$\supseteq S_4$ (3-fold + orthogonal 2-folds)
(A_4, D_4)	$\supseteq S_4$ (automatic)
(A_4, S_3)	$\supseteq S_4$ (automatic)
(D_4, S_3)	$\supseteq S_4$ (automatic)

In particular, for each such pair, any abstract tile P admitting both H_1 and H_2 as non-conjugate subgroups of $\text{Sym}(P)$ satisfies $|\text{Sym}(P)| \geq 24$ and $\text{Sym}(P)$ contains a copy of S_4 acting on a cube.

Proof sketch. Each listed pair either (i) combines a 4-fold and a 3-fold rotation, or (ii) combines a 3-fold rotation with a V_4 subgroup of orthogonal 2-folds, or (iii) already contains A_4 . In case (i) the crystallographic restriction and the classification of finite rotation groups of $\mathbb{R}^3$ force the generated subgroup to be the rotation group of the cube (order 24) or its full symmetry group (order 48); in case (ii) the three mutually orthogonal 2-folds and the 3-fold permuting them generate the same rotation subgroup; case (iii) is immediate since $A_4 \leq S_4$. A finite verification on all unordered pairs of conjugacy classes in T_d (performed computationally; see `scripts/independent_verifier.py` in the public companion repository) confirms no other non-conjugate pair realises a smaller containing group. $\square$

Remark 11.3 (Why this is a screening lemma, not an impossibility theorem). [lemma 11.2](#) is used as a *finite screening device* on orbit decompositions of $n \nmid 24$: whenever a putative decomposition assigns stabilisers in a pair of the table above to distinct orbits, the abstract tile must admit a cubic-or-larger symmetry group, which is incompatible with the finite list of tile shapes compatible with a tetrahedral dissection (verified case-by-case in the obstruction stack, [Table 13](#)). It is *not* a standalone impossibility theorem; the final kill still requires either T_d -maximality ([Section 11.3](#), with the caveat of [remark 11.4](#)) or the Stabiliser Transport Lemma ([lemma 12.2](#)).

11.3 The T_d -maximality argument

Status. The argument of this subsection is **plausible but not fully rigorous**: it relies on the implicit assumption that the cross-orbit congruence $P_1 \rightarrow P_2$ extends to a global isometry of

T , which is not forced by the dissection hypothesis alone. See [remark 11.4](#) immediately below and the Stabiliser Transport Lemma ([lemma 12.2](#), [Section 12](#)) for the correct group-theoretic completion. Decompositions marked (†) in [Table 13](#) are precisely those whose filtering depends on this conditional argument.

For orbit decompositions surviving the crystallographic filter – those with *same-type* stabilisers (e.g. $n = 9 = 3 + 3 + 3$ with all D_4) – the cross-orbit congruence map $P_1 \rightarrow P_2$ must be an isometry of $\mathbb{R}^3$ that is *not* in T_d : T_d permutes pieces within orbits, not between them. Since $T_d = \text{Isom}(T)$ is the full isometry group of T , any such map cannot preserve T globally.

Remark 11.4 (Caveat on [Section 11.3](#)). A congruence $\sigma: P_i \rightarrow P_j$ only needs to map one piece to another; it does not need to preserve T globally. Hence $\sigma \notin T_d$ does not *itself* produce a contradiction: the argument of [Section 11.3](#) is plausible but not fully rigorous, and the Stabiliser Transport Lemma of [Section 12](#) ([lemma 12.2](#)) provides the correct group-theoretic completion.

11.4 Results of the obstruction stack

Systematic enumeration of all integer partitions of n into valid orbit sizes ((6)), followed by stabiliser-type analysis, gives:

Table 13: Non-transitive orbit decompositions filtered for $n \leq 16$, $n \nmid 24$. A dagger (†) indicates decompositions that require the T_d -maximality argument of [Section 11.3](#), plausible but not yet rigorous.

n	Total decomp.	After filter	All filtered?	Primary mechanism
5	6	1	yes	$K \geq S_4$
7	12	3	yes	$K \geq S_4$
9	22	6	yes†	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
10	30	8	yes†	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
11	36	10	yes	$K \geq S_4$
13	58	17	yes	$K \geq S_4$
14	76	21	yes†	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
15	90	26	yes†	$K \geq S_4 + \text{same-type} + T_d\text{-max}$
16	115	33	yes†	$K \geq S_4 + \text{same-type} + T_d\text{-max}$

Four of the nine cases are rigorously filtered by the crystallographic restriction alone (marked without dagger in [Table 13](#)); the remaining five additionally require the T_d -maximality argument ([Section 11.3](#)), which is plausible but not yet fully rigorous – see [remark 11.4](#). We therefore do *not* claim theorem-level impossibility of non- G_{diss} -transitive dissections for $n \leq 16$: what the table establishes is that, within the T_d -invariant category, every tested orbit decomposition is ruled out *up to* the T_d -maximality caveat. This is recorded at level L1+ in [Table 14](#) and motivates (but does not prove) [conjecture 1.2](#).

11.5 Argument chain summary

Table 14: Evidence-level stack of the screening argument against non- G_{diss} -transitive dissections, under the standing hypothesis of T_d -invariance.

Step	Statement	Status
1	Non-transitive $\Rightarrow \geq 2$ orbits, sizes divide 24	Theorem
2	At most one T_d -invariant piece (lemma 11.1)	Theorem
3	Stabiliser action on vertices forces $ V(P) \cap V(T) \in \{0, 4\}$	Proposition (convex)
4	Mixed stabiliser types $\Rightarrow K \geq S_4$ (crystallographic restriction)	Theorem
5	Same-type orbits require non- T_d isometry	Plausible (L1+)
6	S_4 -symmetric bodies cannot tile T for $n \nmid 24$	L1 evidence

12 The rigidity and configurational programme

We now describe the programme that attacks [conjecture 1.2](#) and contains the two principal conditional configurational results of the paper: [conditional theorem 12.16](#) (the $n = 5$ BRH conditional obstruction) and [conditional theorem 12.8](#) ($n = 16$ all-one-face conditional impossibility).

Throughout this section we adopt two standing hypotheses, consistent with all prior literature on congruent simplicial tilings: (i) every piece is convex, and (ii) the dissection is *face-to-face* (every interior 2-face of the piece complex is shared by exactly two pieces).

12.1 Dihedral rigidity programme

A direct attack via dihedral-angle analysis exploits the irrationality of α/π (Conway–Jones [3]). The intended chain was: $\alpha/\pi \notin \mathbb{Q} \Rightarrow$ interior edges carry zero α -coefficient $\Rightarrow$ Dehn localises to the original T -edges $\Rightarrow$ every piece is an α -carrier $\Rightarrow$ every piece touches ≥ 2 of T 's 4 faces $\Rightarrow$ Face-Normal Rigidity forces the gluing isometry $\phi \in T_d \Rightarrow$ transitivity $\Rightarrow n \mid 24$.

Theorem 12.1 (Face-Normal Rigidity). *Let $Q \in O(3)$ be an orthogonal linear map. If Q sends at least two of the four face normals of T (as unit vectors based at the origin) to face normals of T , then Q is the linear part of some element of T_d . Equivalently, for every isometry $\phi(x) = Qx + t$ of $\mathbb{R}^3$ whose linear part Q sends two or more face normals of T to face normals of T , one has $Q \in T_d$ (regarded as a subgroup of $O(3)$ in the standard way). Verified exhaustively on all 144 ordered pairs of face normals.*

However the chain above is broken by two identified gaps:

GAP A (Dehn localisation). The chain assumes all piece dihedral angles have the form $\theta = m\alpha + q\pi$ with $m \in \mathbb{Z}$. This is unjustified for arbitrary cuts: only original T -edges carry α -related angles a priori, and global cancellation $\sum m_i = 0$ at an interior edge does *not* force each m_i to vanish. The naïve “ $m \geq 2 \Rightarrow \theta > \alpha$ ” dichotomy also fails, e.g. $\theta = 2\alpha - \pi/2 \approx 51^\circ < \alpha$.

GAP B (Intrinsic boundary). A face of a piece lying on ∂T is not intrinsically distinguishable from an interior face: the gluing isometry $\phi: P_1 \rightarrow P_2$ could map a boundary face of P_1 to an interior face of P_2 . Without intrinsic distinguishability the Face-Normal Rigidity theorem ([theorem 12.1](#), which *is* valid as a finite test) cannot be invoked to force $\phi \in T_d$.

[theorem 12.1](#) is therefore a correct rigidity tool, but its deployment requires first discharging GAPs A and B. The remainder of [Section 12](#) addresses these gaps.

12.2 Reduction engine and Stabiliser Transport

Following the GAP A/B diagnosis we pivot to a *reduction strategy*: defer the final kill and restrict the shape of any putative counterexample.

Lemma 12.2 (Stabiliser Transport – L2 theorem). *Let $T = P_1 \cup \dots \cup P_n$ be a congruent dissection with congruence group $G = \langle \sigma_{ij} \rangle \leq \text{Isom}(\mathbb{R}^3)$. Then:*

- (a) *stabilisers of congruent pieces are conjugate in $\text{Isom}(\mathbb{R}^3)$;*
- (b) *if $n \nmid 24$ then $G \not\leq T_d$; consequently G contains a non-tetrahedral congruence and the T_d -action on the pieces has ≥ 2 orbits.*

lemma 12.2(a) is standard; (b) follows from the fact that T_d has 30 subgroups in 11 conjugacy classes with index set $\{24/|H|\} = \text{div}(24)$.

Definition 12.3 (Boundary-Normal set). For each face f of the abstract tile P , the *boundary-normal set*

$$\text{BN}(f) := \{ i \in \{0, 1, 2, 3\} : \exists \text{ placement } \phi \text{ with } \phi(f) \subset F_i(T) \}$$

records which T -faces $F_0, \dots, F_3$ the abstract face f can land on.

Conjecture 12.4 (Boundary-Normal Orbits, L1). In a minimal counterexample to [conjecture 1.2](#), the boundary faces of the abstract tile P fall into at most two $\text{Sym}(P)$ -orbits under the 4-normal dictionary $\text{BN}(\cdot)$.

The face-orbit analysis for all 11 subgroup conjugacy classes of T_d partitions $\{0, 1, 2, 3\}$ into progressively finer schemes: e.g. $\mathbb{Z}_3$ acts as $\{0\} \mid \{1, 2, 3\}$, V_4 as $\{0, 1\} \mid \{2, 3\}$, which constrains the boundary structure of any counterexample.

12.3 Spherical Link Program and the Conditional Separation Lemma

The local geometry at each T -vertex V is captured by the *vertex link*, an equilateral spherical triangle Δ_V . Its angles are equal to the dihedral α of T , and its sides equal the face-angle $\pi/3$: both are exactly computable.

Proposition 12.5 (Vertex-link geometry, L2). *The vertex link Δ_V at every vertex of T is equilateral with spherical angle $\alpha = \arccos(1/3)$ and side length $\pi/3$.*

Proof. Apply the spherical law of cosines to the three faces meeting at V : $\cos a = \cos \alpha / (1 - \cos \alpha) = 1/2$, whence $a = \pi/3$. Equality of the three angles is immediate from the symmetry $\text{Sym}_V(T)$. $\square$

Proposition 12.6 (Conditional Separation Lemma, L1+). *Suppose Hypothesis H holds: every interior angle of every link-tile at every T -vertex belongs to $\mathbb{Q}\alpha + \mathbb{Q}\pi$. Then under the Gauss–Bonnet constraint on s -gon tiles Q of Δ_V , the identity separates into*

$$\sum_{\ell} a_{\ell} = \frac{3}{k}, \quad \sum_{\ell} b_{\ell} = s - 2 - \frac{1}{k}, \quad a_{\ell}, b_{\ell} \in \mathbb{Q},$$

where k is the number of pieces meeting at V . In a uniform convex model this forces $k \in \{1, 3, 6, 9, 12, \dots\}$, eliminating $k \in \{2, 4, 5\}$.

Three open gaps prevent [proposition 12.6](#) from attaining theorem status: **GAP C** (Hypothesis H), **GAP D** (mixed vertex types may produce non-congruent link-tiles), **GAP E** (non-convex pieces). Discharge of GAP C is the key conditional step.

12.4 Configurational closure at $n = 16$: the all-one-face residual

[conjecture 12.4](#) targets GAP B by classifying boundary-face orbits abstractly. The same gap can be attacked *configurationally*: enumerate, for fixed piece count n , the candidate “abstract role distributions” of pieces on ∂T and eliminate them one by one.

For $n = 16$ the most constrained distribution places one boundary face of every piece on the *same* single T -face; we call this the *all-one-face* residual.

Setup. Embed T via the rational coordinates

$$v_0 = (1, 1, 1), \quad v_1 = (1, -1, -1), \quad v_2 = (-1, 1, -1), \quad v_3 = (-1, -1, 1), \quad (7)$$

so $|v_i v_j|^2 = 8$ and $V_T = 8/3$. The medial triangle of each T -face decomposes that face into four sub-triangles (*slots*): three corner slots (at the three T -vertices) and one central slot (the medial triangle itself). The conditional theorem uses the following two explicit finite hypotheses to cut the search space:

(H_{roles}) (Role-type assignment). The medial decomposition is the only abstract role-type assignment compatible with face-to-face adjacency: each boundary footprint is either a corner slot or the central slot.

(H_{slot}) (Slot separation). Every piece’s boundary footprint is exactly one of the four slots above; no piece’s footprint mixes slot types.

Both (H_{roles}) and (H_{slot}) are unconditional in the *Coxeter-pure* instance (where every piece is a union of A_3 -orthoschemes) but residual in the general convex instance.

Theorem 12.7 ($n = 16$ all-one-face: Coxeter-pure instance). *There is no face-to-face Coxeter-pure congruent dissection of T into 16 mutually congruent pieces such that every piece has exactly one boundary face on a single common T -face.*

Proof. In the Coxeter-pure instance both (H_{roles}) and (H_{slot}) are automatic (a union of A_3 -orthoschemes has a unique role assignment and a unique boundary-slot type per piece; see [Section 12.4](#)). The conclusion therefore follows from the conditional [conditional theorem 12.8](#) below. $\square$

Conditional Theorem 12.8 ($n = 16$ all-one-face: general convex instance). *Assume (H_{roles}) + (H_{slot}). Let $\{P_1, \dots, P_{16}\}$ be a face-to-face congruent dissection of T into 16 mutually congruent convex polytopes such that every P_i has exactly one boundary face on a single common T -face. Then no such dissection exists.*

The proof assembles three rigidity lemmas, each independently certified by exact rational arithmetic (Python **Fractions**, zero failed tests across the chain).

Lemma 12.9 (Tetrahedrality reduction, C24x). *Every P_i is a tetrahedron with triangular medial base on ∂T and a single interior apex.*

Proof. Two parts.

(I) *Scaled-top family.* No *scaled-top* piece – right prism ($t = 1$), right frustum ($t \in (0, 1)$), or right pyramid with apex above the base centroid ($t = 0$) – fits. The volume constraint $V_{\text{piece}} = V_T/16 = 1/6$ with base area $\sqrt{3}/2$ forces height $h_{\text{piece}} = h_T/4$ (prism) or the frustum-corrected analogue.

- *Layer A (prism).* At $h_{\text{prism}} = \sqrt{3}/9$ the T -cross-section has squared inradius $(121/144)(2/3)$, while the medial triangle has squared circumradius $2/3$; the ratio $144/121 > 1$, so at least one medial vertex sticks out of T .

- *Layer B (frustum)*. For any base point Q the difference between the corner-frustum top and the central-frustum top equals $(1-t)(c_{\text{corner}} - c_{\text{central}})$. Since $|c_{\text{corner}} - c_{\text{central}}|^2 = 2/3 \neq 0$ exactly, face-to-face matching on the shared medial edge fails for every $t \in [0, 1)$.

(II) *Multi-apex residual*. Every convex piece with $k \geq 2$ interior vertices is excluded by two independent rigorous arguments. Let F_1 be the lateral face of a corner- v_j piece carrying the medial edge (m_{jk}, m_{jl}) , and let q be the number of interior vertices of the piece on F_1 .

- **(ARG-1) F_4 parity-pairing** ($k = 2, q = 2$, Type-2 5-vertex polytope). The apex-interior face $F_4 = (v_j, A, B)$ contains v_j and both interior vertices but no midpoint. At vertex v_j exactly three corner- v_j pieces meet, each carrying one such F_4 . Face-to-face matching forces these three faces to pair *among themselves*, but a fixed-point-free involution on a 3-element set is impossible.
- **(ARG-2) Three-plane collapse** (all $k \geq 2, q \geq 2$). Congruence on edge multisets forces every interior vertex X on F_1 of a corner- v_1 piece to satisfy $|v_1 - X|^2 = |m_{kl} - X|^2$, i.e. X lies on the perpendicular bisector plane of (v_1, m_{kl}) . Face-to-face propagation across the three T -faces incident to v_1 yields three such planes:

$$2x - y - z = 1, \quad x - y - 2z = 1, \quad x - 2y - z = 1.$$

The coefficient matrix is non-singular with determinant -4 ; the unique intersection point is

$$P_* = \frac{1}{4}(1, -1, -1),$$

a rational interior point of T with barycentric coordinates $(3, 7, 3, 3)/16$. Every interior vertex on F_1 collapses to P_* , contradicting $q \geq 2$ distinct vertices.

- The case $q = 1$ (triangular F_1) reduces to [lemma 12.10](#) below via face-by-face apex collapse along the three medial edges.

This completes the proof. □

Lemma 12.10 (Apex collapse, C24w). *Conditional on [lemma 12.9](#), every congruent piece has a single interior apex at perpendicular height $h_T/4$ above its medial base, and four rigidity layers force all 16 apices to coincide at the centroid $G_T = (0, 0, 0)$:*

- (L_1) Volume forces apex height $= h_T/4$.
- (L_2) Same-face medial-edge matching forces the apex of the corner- v_j piece on face F_i to equal the apex of the central piece on F_i .
- (L_3) T -edge matching propagates the apex across faces, forcing all four central apices to coincide at a common point Q .
- (L_4) Q is at perpendicular distance $h_T/4$ from every face; the 3-of-4 face-normal linear system has unique solution $Q = G_T$.

Lemma 12.11 (Edge-multiset incongruence at G_T , C24v). *At $Q = G_T$, the corner piece has squared-edge multiset $\{1, 1, 2, 2, 2, 3\}$ and the central piece has squared-edge multiset $\{1, 1, 1, 2, 2, 2\}$. These multisets differ; hence the corner and central pieces are not congruent.*

Proof of conditional theorem 12.8. Combine [lemma 12.9](#), [lemma 12.10](#), and [lemma 12.11](#) with the medial-base reduction encoded in $(H_{\text{roles}}) + (H_{\text{slot}})$: the first forces tetrahedrality; the second collapses all apices to G_T ; the third produces a direct violation of piece congruence. □

Remark 12.12 (Scope of [theorem 12.7](#) and [conditional theorem 12.8](#)). In the Coxeter-pure instance $(H_{\text{roles}}) + (H_{\text{slot}})$ are automatic, so [theorem 12.7](#) is the L3 (unconditional) specialisation of the L2 conditional statement [conditional theorem 12.8](#). In the general convex instance $(H_{\text{roles}}) + (H_{\text{slot}})$ remain hypotheses: an unconditional proof of them in that setting is not supplied here. Thus [conditional theorem 12.8](#) is at evidence level L2.

Remark 12.13 (Convexity hypothesis). The convexity of the pieces is essential to [lemma 12.9](#): without it, multi-apex non-convex pieces are not reached by the perpendicular-bisector argument (ARG-2). Discharging convexity is natural future work; the present theorem is, as stated, in the convex category.

Remark 12.14 (Why $n = 16$?). $n = 16$ divides neither 24 nor 48. Note that closing the full $n = 16$ case *would* advance [conjecture 1.1](#), since $16 \nmid 24$ means that the conjecture predicts no dissection should exist. However, [lemma 12.9](#) closes only the *all-one-face residual* configuration at $n = 16$, not the entire case. This particular configuration was historically the most resistant residual of the role-distribution table in the programme described in [Section 12.2](#). The same C24 machinery is structurally portable to the other non-divisors $n \in \{20, 28, \dots\}$ and gives a template for the configurational attack on GAP B.

12.5 The $n = 5$ case: a conditional boundary-role obstruction

The previous public reduction of arbitrary convex face-to-face five-piece dissections to one accidental-even metric residual is not established. Its boundary-role step requires an additional hypothesis: congruent copies need not expose the same facets of the common reference polytope. We therefore replace that reduction by the conditional result below. It is a complete obstruction under an explicit boundary-role homogeneity condition (BRH), while the unrestricted case with different boundary roles remains open.

Definition 12.15 (Boundary-role homogeneity (BRH)). Let $P_1, \dots, P_5$ be full-dimensional congruent convex polytopes with union T and pairwise disjoint interiors. BRH holds if there are a reference copy P , Euclidean isometries γ_i with $\gamma_i(P) = P_i$, and a fixed subset B of facets of P such that $\gamma_i(B)$ is exactly the set of facets of P_i contained in ∂T . The set B is the common boundary-role set. BRH requires congruences preserving these designated facets; ordinary congruence or equality of touch counts does not by itself provide such congruences.

Conditional Theorem 12.16 ($n = 5$ under BRH). *No partition of a regular tetrahedron into five full-dimensional congruent convex polytopes satisfies BRH. In particular, five-piece impossibility is proved conditionally on BRH, but no unconditional $n = 5$ impossibility is claimed for congruent copies with different boundary roles.*

Proof. Let A be the area of one face of T , and let $t = |B|$. The boundary footprints cover the four faces of T up to sets of area zero. Indeed, two full-dimensional convex pieces contained in T cannot share a relatively open subset of an ambient face without overlapping interiors, so their boundary footprints can overlap only in area-zero sets. Since BRH identifies the exposed facets in every copy, their total area is the same in each piece; hence the common exposed area is

$$E = \frac{4A}{5}. \quad (8)$$

Every exposed facet has area at most E , because it is a positive summand of that total. Thus every face of T needs at least two boundary footprints. Convexity gives at most one exposed facet on each ambient face, so $t \leq 4$.

If $t = 0$, there is no boundary coverage. If $t = 1$, there are only five footprints in total, whereas four faces require at least eight. Suppose $t = 2$, and write the two facet areas as $a \leq b$. Normalize $A = 1$, so $a + b = 4/5$. There are ten footprints, so at least one face has exactly two footprints and another has at least three. A two-footprint face has area $2a$, $a + b$, or $2b$. The

first two are less than 1, so $2b = 1$ and $(a, b) = (3/10, 1/2)$. Three footprints then have one of the four sums $9/10, 11/10, 13/10, 15/10$, while four or more have area at least $4a = 6/5$. No face with at least three footprints can therefore have area 1, a contradiction.

It remains to consider $t = 3$ or $t = 4$. Center T at the origin, let $\nu_0, \dots, \nu_3$ be its outward unit face normals, and let $h > 0$ be the common support number, so that

$$T = \{x \in \mathbb{R}^3 : \nu_j \cdot x \leq h \text{ for } j = 0, \dots, 3\}.$$

They satisfy $\nu_0 + \nu_1 + \nu_2 + \nu_3 = 0$, and every three are linearly independent. Replace the reference tile by P_1 and set $\phi_i = \gamma_i \gamma_1^{-1}$ for $i = 1, \dots, 5$. Write $\phi_i(x) = Q_i x + u_i$. The map ϕ_i sends three exposed facets of P_1 to exposed facets of P_i . At an exposed facet the piece's outward normal is the *positive* ambient outward normal $+\nu_j$ (not $-\nu_j$), and isometries transport outward normals. Thus Q_i sends three distinct tetrahedral face normals to three distinct tetrahedral face normals. The zero-sum identity forces the fourth normal to be sent to the fourth one as well. Hence Q_i permutes the four normals.

For a matched facet on $\nu_j \cdot x = h$, the image facet lies on the corresponding ambient support plane. For x on that facet,

$$h = (Q_i \nu_j) \cdot (Q_i x + u_i) = h + (Q_i \nu_j) \cdot u_i.$$

This makes u_i perpendicular to three independent normals, so $u_i = 0$. Every ϕ_i is consequently an element of the full tetrahedral symmetry group H . Its order is 24: each permutation of the four vertices of a regular tetrahedron extends uniquely to an orthogonal symmetry.

For completeness, the required orbit count does not assume that H permutes the partition. Choose a point x in the interior of T outside the finite union of all fixed-point sets of nonidentity elements of H and all inverse images under H of the piece boundaries. Such a point exists because this finite union has three-dimensional measure zero. Then $O = Hx$ has 24 points and meets no piece boundary. Since $P_i = \phi_i(P_1)$ and $\phi_i \in H$, $|O \cap P_i| = |O \cap P_1|$ for every i . If this common number is m , partitioning O gives $24 = 5m$, impossible. This proves the conditional theorem. $\square$

The proof does not require the partition to be face-to-face. Its restriction is BRH: it does not establish a common boundary-role set for arbitrary congruent copies. Establishing that property for a specified class, or excluding mixed boundary roles directly, remains a separate problem.

Observation 12.17 (Exact bookkeeping for a common exposed-area list). Assume only that each piece has the same list of positive exposed-facet areas $m = (m_1, \dots, m_t)$, without requiring the facet identifications to be induced by congruences. Boundary coverage again gives $\sum m_r = 4A/5$. Normalize $A = 60$, hence $E = 48$; this does not require individual areas to be integers. Label equal-area occurrences arbitrarily. Let N_{jr} count occurrences of label r on ambient face j . Necessary equations are

$$Nm = 60(1, 1, 1, 1)^T, \quad m_r > 0,$$

with row sums k_j and column sums 5. Each k_j lies between 2 and 5, and $\sum k_j = 5t$. For $t = 2, 3$ the table lists every possible sorted k and every positive area solution of these count equations. The script `scripts/n5_real_area_certificate.py` enumerates all 3461 matrices with these margins and solves the rational linear systems exactly. No consistent rank-deficient system occurs, so no real solution family is lost by listing isolated rational solutions.

t	k	positive sorted area multisets
2	(4, 2, 2, 2)	none
2	(3, 3, 2, 2)	none
3	(5, 5, 3, 2)	none
3	(5, 4, 4, 2)	(6, 12, 30)
3	(5, 4, 3, 3)	(6, 15, 27), (6, 18, 24), (48/5, 84/5, 108/5), (12, 12, 24)
3	(4, 4, 4, 3)	(6, 18, 24), (12, 12, 24), (12, 16, 20)

There are six distinct positive multisets and eight residual occurrences. The rational multiset $(48/5, 84/5, 108/5)$ is genuine at the incidence level; one aligned witness is

$$N = \begin{pmatrix} 4 & 0 & 1 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}, \quad m = \left(\frac{48}{5}, \frac{84}{5}, \frac{108}{5} \right)^\top.$$

Its row sums are $(5, 4, 3, 3)$ and every weighted row sum is 60. The calculation concerns areas and incidence only; no matrix here is a geometric dissection certificate.

Observation 12.18 (A local algebraic area model). The following identity retains a local calculation from the NN first-factor model, with all algebraic assumptions explicit. For real x, y, z, L with $L \neq 0$, define

$$\begin{aligned} Q_p &= x^2 - xy + y^2, & Q_r &= z^2 - z + 1, \\ B &= x^2 + xy + y^2 - (z^2 + z + 1), & D &= xy - z, \\ E_\lambda &= Q_r - 12L^2Q_p, & E_{\text{axis}} &= Q_p(4 - (6L - 1)^2) - 12B, \\ A_{\text{floor}} &= D + 2LQ_p + \frac{Q_r}{6L}, & A_{\text{total}} &= A_{\text{floor}} + B + D. \end{aligned}$$

Then the identity

$$24L \left(A_{\text{floor}} - \frac{3}{4}A_{\text{total}} \right) = 2LE_{\text{axis}} + (1 - 6L)E_\lambda$$

holds identically; substituting $B = Q_p - Q_r + 2D$ verifies it directly. If $E_\lambda = E_{\text{axis}} = 0$ and $A_{\text{total}} > 0$, it follows that $A_{\text{floor}}/A_{\text{total}} = 3/4$. The identity is checked symbolically by `scripts/n5_nn_identity_certificate.py`. If these quantities describe one exposed facet and the total exposed area of a piece in the $t = 3$ common-area model of [observation 12.17](#), they are incompatible with its maximum facet-area ratio $30/48 = 5/8$. This last conclusion requires that geometric identification. No assertion that arbitrary pieces, all NN branches, or mixed boundary roles enter this algebraic model is made.

Remark 12.19 (Withdrawn metric shortcuts). The earlier $f = 5$ triangular-prism exclusions are not used here. A triangular-prism face lattice does not imply parallelism of the two triangular faces. In affine coordinates with $T = \{x_i \geq 0, \sum_i x_i \leq 1\}$, take $0 < a < \min_i b_i$ and $\max_i b_i \leq 1$. The polytope

$$x_i \geq 0, \quad x_1 + x_2 + x_3 \geq a, \quad \sum_i x_i/b_i \leq 1$$

has vertices ae_i, b_ie_i and the triangular-prism face lattice; its two triangular facets are nonparallel when the b_i are not all equal. Likewise, a setwise facet correspondence does not by itself imply pointwise fixing or an orientation-reversing reflection. The old H -cocycle closure is therefore not imported into the present theorem.

Remark 12.20 (Current $n = 5$ boundary). The active result is exactly [Conditional Theorem 12.16](#), the conditional BRH obstruction, together with the finite area and local-chart observations above. The unrestricted convex face-to-face five-piece question remains open. The historical phase-26 scripts and result files are retained for auditability but are no longer an active proof chain. The reproducer runs the BRH arithmetic checks, the area enumeration, the local algebraic identity, and the auxiliary atlas checks described in the repository README. The BRH proof is a written geometric argument, not an inference from passing test counts.

12.6 Summary of the programme

[conjecture 1.2](#) is equivalent to [conjecture 1.1](#) combined with [Corollary 7.4](#). Its current evidence level is L1+, supported by

- one L2 theorem ([lemma 12.2](#), Stabiliser Transport);
- one L1+ proposition ([proposition 12.6](#), Conditional Separation, conditional on Hypothesis H);
- one conjecture at L1 ([conjecture 12.4](#), Boundary-Normal Orbits);
- a Spherical Link Program constraining pieces-per-vertex to $k \in \{1, 3, 6, \dots\}$ within a uniform convex model under Hypothesis H;
- the dihedral rigidity programme ([Section 12.1](#)) reducing [conjecture 1.2](#) to GAPs A, B;
- the $n = 16$ all-one-face conditional reduction ([theorem 12.7](#) and [conditional theorem 12.8](#)), L3 in the Coxeter-pure case and L2 (conditional on $(H_{\text{roles}}) + (H_{\text{slot}})$) in the general convex case;
- and the conditional $n = 5$ BRH obstruction ([conditional theorem 12.16](#)). The unrestricted case with different boundary roles remains open; exact area enumeration and the local NN identity provide diagnostics but do not close that gap.

13 Further questions

Question 13.1 (Hinged dissections). Can any of the eight known dissections be realised as a physical hinged mechanism? Edge-hinges (1-DOF) are natural candidates for S_2 and S_4 , vertex-hinges (3-DOF) for the dense configurations S_{24} . No results are known in this direction. Demaine’s universality theorems on scissors-congruent polyhedra [5] apply to pairs of polyhedra, not to partitions of a single polyhedron; their adaptation to the present setting is open.

Question 13.2 (Higher-dimensional extension). For the regular d -simplex with symmetry group S_{d+1} of order $(d+1)!$, does n admit a congruent dissection if and only if $n \mid (d+1)!$? The answer is *no* for $d = 2$: the equilateral triangle admits congruent dissection into any $n = k^2$ pieces via self-similar subdivision. The tetrahedron ($d = 3$) may be the critical dimension at which the Dehn invariant first constrains the admissible n ; this should be a theorem for $d = 3$, with modifications for $d \geq 4$.

Question 13.3 (Burnside ring extensions). The Burnside ring framework of [Section 9](#) opens questions in *equivariant K-theory* of simplicial complexes and connections to the representation ring $R(T_d)$. The relationship between the mark table and the character table of S_4 may encode deeper number-theoretic structure.

A Computational reproduction sketch

The released metric, Dehn, stabiliser, and SAT-style certificates use exact rational arithmetic, including Python’s built-in `Fractions` module. We include below a minimal self-contained reference implementation for the Dehn-invariant verification at $n = 24$ (orthoscheme). The public replay suite for [conditional theorem 12.16](#), together with the auxiliary atlas checks, is available in `scripts/` and `results/` of the companion repository identified in [Section 1](#). The active $n = 5$ script certifies only the finite arithmetic checks used by the BRH proof; historical phase-26 reduction scripts are not part of that active chain. As stated there, the two general-convex hypotheses of [conditional theorem 12.8](#) are not computationally discharged by this public package.

Listing 1: Minimal SymPy verification of the Dehn scaling $D(S_{24}) = D(T)/24$ for the orthoscheme piece.

```

1 import sympy as sp
2 from sympy import Rational, Point3D, sqrt
3 # Regular tetrahedron T with edge length 1
4 A = Point3D(0, 0, 0)
5 B = Point3D(1, 0, 0)
6 C = Point3D(Rational(1, 2), sqrt(3)/2, 0)
7 D = Point3D(Rational(1, 2), sqrt(3)/6, sqrt(6)/3)
8 # Canonical orthoscheme piece S_24
9 M_AB = (A + B) / 2
10 G_ABC = (A + B + C) / 3
11 G_T = (A + B + C + D) / 4
12 S24 = [A, M_AB, G_ABC, G_T]
13 # Edge enumeration, dihedral-angle computation via dot products of
14 # outward face normals, and Dehn tensor product in R (x)_Z (R/pi Q)
15 # follow directly; total runtime < 1 s.

```

B Evidence-level glossary

- L0** *Heuristic.* A claim supported by plausibility or partial checks but without a formal proof attempt.
- L1** *Partial evidence.* A claim supported by substantial evidence (e.g. computational enumeration up to a fixed bound) but lacking a closed proof.
- L1+** *Strong partial evidence.* A claim supported by a partial proof together with the explicit identification of the remaining residual gap(s); typically a reduction to a named sub-lemma.
- L2** *Conditional theorem.* A rigorously proved implication whose conclusion depends on an explicit, named auxiliary hypothesis (e.g. convexity, face-to-face, or a named hypothesis H).
- L2+** *Finite-check conditional theorem.* Like L2, but each standing auxiliary hypothesis is a finite, explicitly verifiable condition on a candidate configuration (e.g. a cocycle identity, an orbit table, or a single-piece combinatorial type exclusion).
- L3** *Unconditional theorem.* A fully proved statement with no auxiliary hypothesis beyond those of the ambient setting (convex face-to-face congruent dissection).

C The 24-cell and quaternionic action

This appendix provides a *secondary* algebraic viewpoint. The core results of the paper (Section 3–Section 9, Section 10–Section 11) do not depend on it; it serves as a partial independent verification of the dissection inventory through the quaternionic / 24-cell correspondence, and may be skipped on first reading.

C.1 The binary tetrahedral group $2T$

The binary tetrahedral group $2T \subset S^3 \subset \mathbb{H}$ consists of 24 unit quaternions: the 8 *type-1* units $\{\pm 1, \pm i, \pm j, \pm k\}$ and the 16 *type-2* units $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$. These are precisely the 24 *Hurwitz units*, the vertices of the regular 24-cell inscribed in S^3 . The group has seven conjugacy classes of sizes $(1, 6, 1, 4, 4, 4, 4)$. Its centre is $\{\pm 1\}$, and

$$1 \longrightarrow \{\pm 1\} \longrightarrow 2T \xrightarrow{\pi} A_4 \longrightarrow 1 \quad (9)$$

is exact.

Table 15: Subgroups of $2T$ and dissection matches.

$2T$ -subgroup	$ H $	$[2T : H]$	$\pi(H) \leq A_4$	$ \pi(H) $	Dissection
$\{e\}$	1	24	$\{e\}$	1	S_{24}
$\mathbb{Z}_2$ (centre)	2	12	$\{e\}$	1	S_{12}
$\mathbb{Z}_3$	3	8	$\mathbb{Z}_3$	3	S_8
$\mathbb{Z}_4$	4	6	$\mathbb{Z}_2$	2	S_6
$\mathbb{Z}_6$	6	4	$\mathbb{Z}_3$	3	S_4
Q_8	8	3	V_4	4	S_3
$2T$	24	1	A_4	12	S_1

C.2 Triple bijection

The quaternionic conjugation action $v \mapsto qv\bar{q}$ on pure quaternions v yields a canonical triple bijection

$$\underbrace{24 \text{ Hurwitz units}}_{2T} \longleftrightarrow \underbrace{24 \text{ permutations}}_{T_d \cong S_4} \longleftrightarrow \underbrace{24 \text{ orthoschemes}}_{\text{flags}(T)}. \quad (10)$$

Proper rotations give $24 \twoheadrightarrow 12$ even permutations (A_4) with kernel $\{\pm 1\}$; composition with a fixed involution σ recovers the 12 odd permutations ($S_4 \setminus A_4$); and each of the 24 orthoschemes in the barycentric subdivision of T is hit exactly once.

C.3 Subgroup lattice and the $n = 2$ gap

The subgroup lattice of $2T$ maps under π to the subgroup lattice of $A_4 \leq S_4$. Seven of the eight dissection numbers are recovered directly from coset spaces $2T/H$: $\{1, 3, 4, 6, 8, 12, 24\}$ (Table 15). The missing case $n = 2$ arises because $2T$ has no subgroup of order 12; indeed π has image $A_4 \leq S_4$ of index 2, and all $|\pi(H)|$ divide 12. The $n = 2$ dissection is captured by the S_4 -based Burnside analysis (Section 9) via stabiliser A_4 ; see Table 15.

References

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